

Research Paper

Effect of reentrant cavities on the thermal performance of a pulsating heat pipe

Woogyong Kim, Sung Jin Kim*

Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology, 291 Daehak-ro, Daejeon 34141, Republic of Korea

HIGHLIGHTS

- Flow characteristics of the MPHP with reentrant cavities are investigated.
- Reentrant cavities in the MPHPs are shown to promote nucleation and early startup.
- The thermal resistance of the MPHP with reentrant cavities is decreased by up to 57%.
- A single MPHP incorporating various sizes of reentrant cavities is proposed.

A B S T R A C T

This study is performed to investigate the effect of the size of reentrant cavities on the thermal performance of a micro pulsating heat pipe (MPHP). The flow and thermal characteristics of the MPHPs with each MPHP having a different size of reentrant cavities, along with a MPHP without reentrant cavities are experimentally obtained and compared. Silicon-based MPHPs with and without reentrant-type artificial cavities inside the channels are fabricated using MEMS techniques. The MPHPs have rectangular channels which are engraved on a silicon wafer with a hydraulic diameter of 667 μm . Ethanol is used as the working fluid. To allow for flow visualization, the etched micro-channels are covered with a transparent glass. Each MPHP with reentrant cavities has reentrant cavities of one size, which are either 10, 20, 30, or 40 μm , respectively. Reentrant cavities in the MPHPs are shown to promote nucleation and early startup. Furthermore, the thermal resistance of the MPHP with reentrant cavities is decreased by up to 57%. As the size of the reentrant cavities increases, lower input power is required for startup, and the MPHP with the largest cavities (40 μm) shows the earliest startup. On the contrary, as the size of the reentrant cavities decreases, a reduction of the thermal resistance of the MPHPs is maintained to higher input power, and the MPHP with the smallest cavities (10 μm) shows the lowest thermal resistance at high input power ($> 12 \text{ W}$). Finally, a single MPHP incorporating various sizes of reentrant cavities (10, 20, 30, and 40 μm) is shown to exhibit extended operating range and enhanced thermal performance simultaneously.

1. Introduction

Due to continuous demand for faster and smaller microelectronic systems, heat flux in electronics has been ever increasing. Therefore, effective thermal design for electronic components and systems is a critical issue. A conventional heat pipe, which is a widely used two-phase cooling device, is difficult to change its shape because of the wick structure. A pulsating heat pipe (PHP), which was proposed by Akachi et al. [1], is an alternative to overcoming this limitation. To increase the thermal performance of a PHP, many novel designs have been proposed. Kwon and Kim [2] employed a dual diameter channel to

improve the thermal performance of a PHP. Inoue et al. [3] suggested the use of wires to enhance the thermal performance of a PHP in horizontal mode. Among the novel designs, it was claimed that nucleation provides a large driving force for the oscillating motions of the working fluid in a PHP [4–6]. Senjaya et al. [5] conducted a numerical simulation and presented that a PHP with nucleation sites has a larger heat transfer rate compared to a PHP without nucleation sites for a given temperature difference. Therefore, promoting nucleation in a PHP may enhance the thermal performance of a PHP.

Generally, artificial cavities are widely used to promote nucleation by reducing the wall superheat required for the onset of nucleate

* Corresponding author.

E-mail address: sungjinkim@kaist.ac.kr (S.J. Kim).

Nomenclature

d_{cavity}	size of mouth cavity [m]
$\dot{q}$	heat flux [W/m ²]
k	thermal conductivity [W/m·K]
θ	contact angle [rad]
T	temperature [K]
h_{LV}	latent heat of vaporization [KJ/Kg]
k_{eff}	effective thermal conductivity [W/m·K]
Q_{in}	inlet input power [W]
R_{th}	thermal resistance [K/W]
$\bar{T}$	average temperature [K]

Greek symbols

σ	surface tension
ρ	density

Subscripts, Superscripts

SAT	saturated state
ONB	onset of nucleate boiling
L	liquid
V	vapor
e	evaporator
c	condenser

boiling (ONB) [7–10]. However, there are few studies about what effect artificial cavities have on the thermal performance of a PHP. Qu and Ma [11] conducted a theoretical analysis and suggested that startup can be improved by applying artificial cavities to a PHP. However, their results were not experimentally validated. Moreover, their results are limited to startup of a PHP, so it is not clear how artificial cavities change the thermal performance of a PHP. Technical papers on boiling experiment reported that the heat transfer rate can be deteriorated depending on the geometry of artificial cavities [12–14]. In particular, the effect of artificial cavities on the heat transfer rate is greatly influenced by the size of artificial cavities [13–16]. Owing to these reasons, choosing an appropriate size of artificial cavities is important to improve the thermal performance of a PHP. Therefore, it is necessary to investigate the effect of the size of artificial cavities on the thermal performance of a PHP.

The purpose of the present study is to evaluate the effect of the size of reentrant cavities on the thermal performance of a PHP. Using MEMS techniques, five-turn closed-loop MPHPs with and without reentrant-type artificial cavities are fabricated. The channels are etched onto a silicon wafer with a hydraulic diameter of 667 μm and the etched silicon wafer is covered with Pyrex glass to allow for flow visualization. Reentrant cavities are engraved along the side of the channel walls. Four different MPHPs with four different sizes of reentrant cavities are fabricated. Each of the four MPHPs with reentrant cavities has reentrant cavities of just one size, which are 10, 20, 30, and 40 μm , respectively. The flow and thermal characteristics of these MPHPs are compared using high-speed photography and thermometry. Finally, based on the experimental results, a new design guideline for enhancing the thermal performance of the MPHP is proposed for practical implementation of a PHP with reentrant cavities.

2. Experiments

2.1. Fabrication of MPHP with reentrant cavities

Five-turn closed-loop MPHPs with artificial cavities were fabricated, and the artificial cavities were engraved along the side of the channel walls simultaneously using MEMS techniques. The MPHPs had rectangular channels which were etched onto a silicon wafer with a thickness of 1 mm using deep reactive ion etching (DRIE) process. The width and the height of the micro-channels were 1 mm and 0.5 mm, respectively. To evaluate the effect of the size of artificial cavities on the flow and thermal characteristics of the MPHPs, several sets of the MPHPs with artificial cavities of different sizes were fabricated, as summarized in Table 1 and shown in Fig. 1. The shape of the artificial cavities was

chosen to be a reentrant-type which is widely used for pool boiling experiments [17], flow boiling in micro-channels [18,19], heat pipes [20,21], and commercial heat exchangers [14]. Other geometrical parameters, except for the size of the mouth of the cavity, were maintained at the same proportions. To minimize the flow resistance, reentrant cavities were engraved only in the evaporator section and spaced 500 μm apart. Top view images of the engraved reentrant cavities are shown in Fig. 2. The MPHP without reentrant cavities, which is a plain MPHP, was fabricated to compare its flow and thermal characteristics to those with reentrant cavities. The etched micro-channels were covered with Pyrex glass with a thickness of 1 mm using anodic bonding. The MPHPs had a length of 50 mm, width of 18.5 mm, and thickness of 2 mm, as shown in Fig. 3. The length of the condenser, the adiabatic, and the evaporator section was set to 15 mm, 25 mm, and 10 mm, respectively. Two holes were fabricated for charging the working fluid. Degassing and charging of the working fluid were performed, and the details of these processes can be found in Youn and Kim [22]. Ethanol was used as the working fluid, and the filling ratio was set to 50%. The experimental parameters used in the present study are summarized in Table 2.

Table 1
Specific dimensions of reentrant cavities.

	C10	C20	C30	C40
Mouth (X) [μm]	10	20	30	40
Y [μm]	20	20	20	20
Z [μm]	25	50	75	100

Table 2
Experimental parameters.

Size of cavity [μm]	Hydraulic diameter [μm]	Working fluid	Filling ratio	Input power [W]	Inclination angle (°)
40 30 20 10 Plain	667	Absolute ethanol	50%	2 ~ (Increment:1)	90 (Vertical)

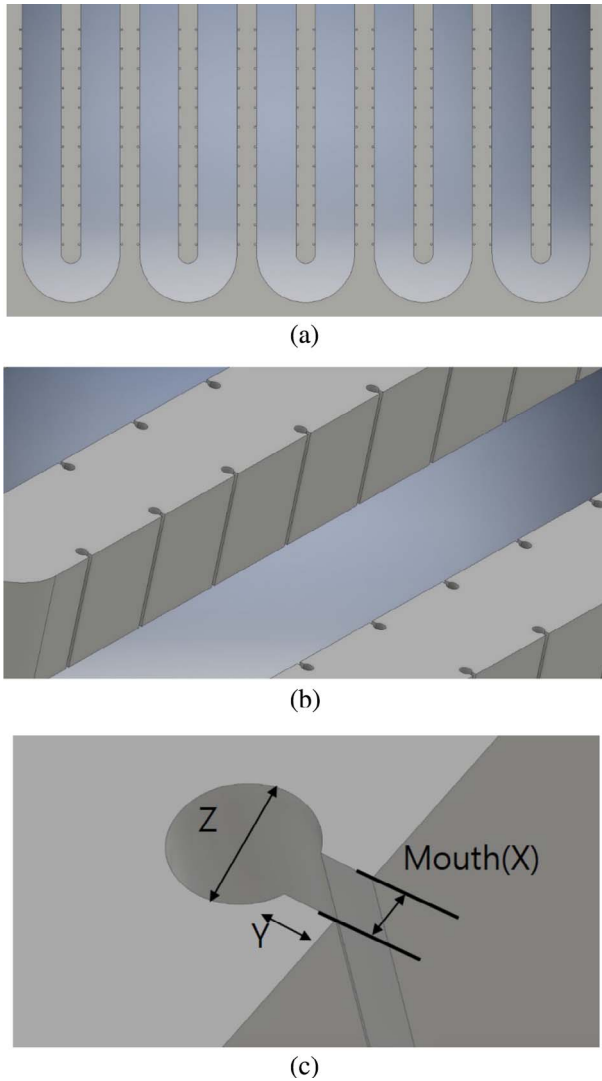


Fig. 1. Schematic of reentrant cavities in the MPHP: (a) Top view, (b) isometric view, and (c) enlarged isometric view.

2.2. Experimental setup

Fig. 4 shows a schematic diagram of the experimental setup used in this study. To minimize the heat loss from the MPHPs to the environment, every experiment was performed in a vacuum chamber (10^{-2} – 10^{-3} Torr). For a thin film heater, nichrome (Ni:Cr = 80:20 wt%, I-Nexus, Inc.) was evenly deposited on the silicon surface of the evaporator section by using a DC magnetron sputtering system. (A-TECH System). The thin film heater was heated by Joule heating using DC power supply (N5772A, Agilent technologies) and input power supplied to the thin film heater was increased from 2 W by increments of 1 W until the average temperature of the evaporator section reached 120 °C. The condenser section was cooled by a copper block into which the coolant, water, was supplied by a bath circulator (RW-0525G, JEIO TECH). The inlet temperature of the coolant was constantly maintained at 20 °C. The temperature of the outer wall of the MPHPs was measured using K-type thermocouples (K-type, OMEGA) attached to the silicon surface of the MPHPs with silver epoxy (A-200). The accuracy of the thermocouples was calibrated to be within $\pm 0.4\%$. The measured

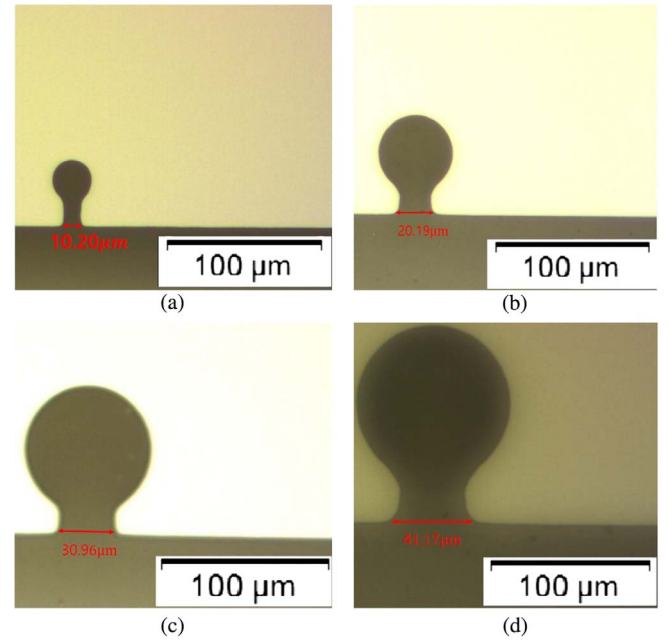


Fig. 2. Top view images of fabricated reentrant cavities: (a) C10, (b) C20, (c) C30, and (d) C40.

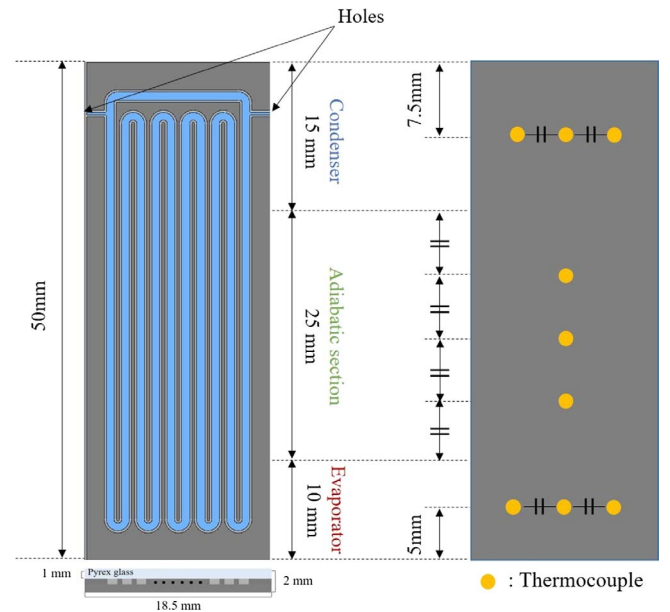


Fig. 3. Schematic and dimensions of the fabricated MPHP.

temperature data were recorded by the data acquisition system (34980A, Agilent technologies) and saved on a computer. Every experiment was conducted in a vertical orientation with bottom heating mode. Then, a high-speed camera (M320S, Phantom) was used to observe the flow characteristics of the MPHPs with and without reentrant cavities.

To evaluate the thermal performance of the MPHPs with varying input power, the thermal resistance of the MPHPs, R_{th} , is defined as

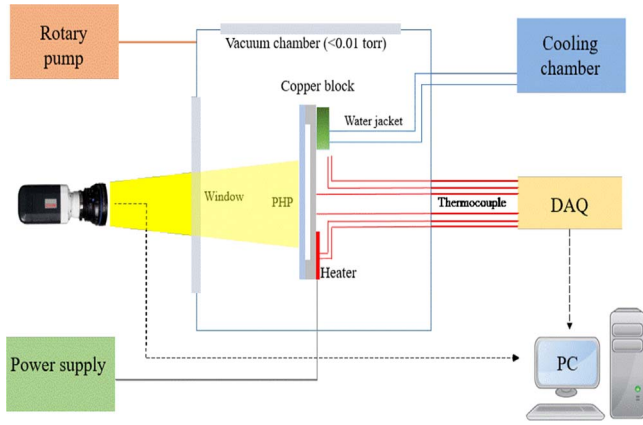


Fig. 4. Schematic of experimental setup.

Table 3

Working range and accuracy of experimental instruments.

Instruments	Working range	Accuracy
DC power supply (N5772A)	0–60 V	±1%
Bath circulator (RW-0525G)	–25 to 150°C	±0.05°C
K-type thermocouple	–100 to 1200°C	±1°C

follows:

$$R_{th} = \frac{\bar{T}_e - \bar{T}_c}{Q_{in}} \quad (1)$$

where $\bar{T}_e$, $\bar{T}_c$, and Q_{in} are the average temperatures of the evaporator and the condenser sections, and input power, respectively. The average temperatures for the thermal resistance are evaluated using an arithmetic mean of temperatures, which are taken 20 min after the MPHPs enter steady normal operation. The power supplied to the thin film heater (Q_{in}) is calculated as follows:

$$Q_{input} = V \times I \quad (2)$$

where V and I are electric voltage and electric current, respectively. Every experimental uncertainty for calculating thermal resistances is estimated using the method suggested by Holman [23]. The working range and accuracy of every instrument are shown in Table 3. The experimental uncertainties are estimated with a 95% confidence interval, and overall uncertainties for calculating the thermal resistance are less than 6%.

3. Results and discussion

3.1. Experimental results

3.1.1. Thermal resistance of MPHP with reentrant cavities

Fig. 5 shows the thermal resistance of the MPHPs with reentrant cavities and the plain MPHP at various input powers. From flow visualization, the operating characteristics of the MPHP with reentrant cavities can be classified into three regimes depending on the thermal resistance: Startup regime (Low input power), cavity-effective regime (Medium input power), and cavity-ineffective regime (High input power). Fig. 6 presents the three regimes of the MPHP with reentrant cavities. The range of each regime of the MPHPs with reentrant cavities

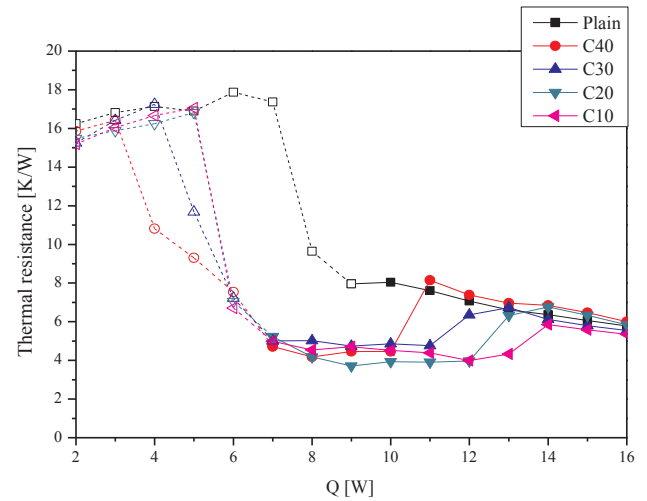


Fig. 5. The thermal resistance of the MPHPs with reentrant cavities and the plain MPHP.

is slightly shifted according to the size of the reentrant cavities. The startup regime includes a non-operating region and a region of unstable operation of the MPHP. Before startup, the MPHP only transfers heat through conduction without any fluid motion. Therefore, the thermal resistance is relatively high. In this situation, the temperature of the evaporator section is higher than that in normal operation. In the case of the plain MPHP, the temperature of the evaporator section increases over 100 °C, which is a very dangerous situation from the view point of practical application. Even after startup, unstable oscillation is observed in the startup regime. As time goes on, an intermittent stopover phenomenon is observed, which means the movement of the working fluid stops at certain times. Then, the working fluid moves again, and such a process is repeated throughout the startup regime. For these reasons, the startup regime should be distinguished from stable normal operation. Thus, the thermal resistance of the MPHPs at the startup regime is expressed with empty symbols.

In the startup regime, every MPHP with reentrant cavities shows early startup compared to the plain MPHP. Fig. 7 shows a startup input power, which is input power required for startup, for different reentrant cavity sizes. Earlier startup is observed for the MPHP with larger reentrant cavities. In the case of C40, the earliest startup is observed among the MPHPs with reentrant cavities.

As the input power increases, the MPHP enters stable normal operation. During normal operation of the MPHPs, the liquid slugs and the vapor plugs oscillate more rapidly compared to conventional PHPs. In the cavity-effective regime, regardless of the size of the reentrant cavities, the thermal performance of the MPHP with reentrant cavities is enhanced significantly compared to the plain MPHP. The MPHP with reentrant cavities have up to 57% lower thermal resistance than the plain MPHP. Fig. 8 presents the temperatures of the evaporator section of the MPHP with reentrant cavities and the plain MPHP at 9 W. When it comes to the temperatures of the evaporator section of the MPHP with reentrant cavities, temperatures during normal operation and startup decrease 40 °C more than compared to the case of the plain MPHP. From these results, it can be inferred that reentrant cavities inside the MPHPs reduce wall superheat at the ONB. However, as the input power further increases, the MPHP with reentrant cavities moves into the cavity-ineffective regime, and the thermal resistance of the MPHPs with reentrant cavities and the plain MPHP become similar. Contrary to the startup regime, as the size of the reentrant cavities increases, the thermal resistance of the MPHPs with reentrant cavities

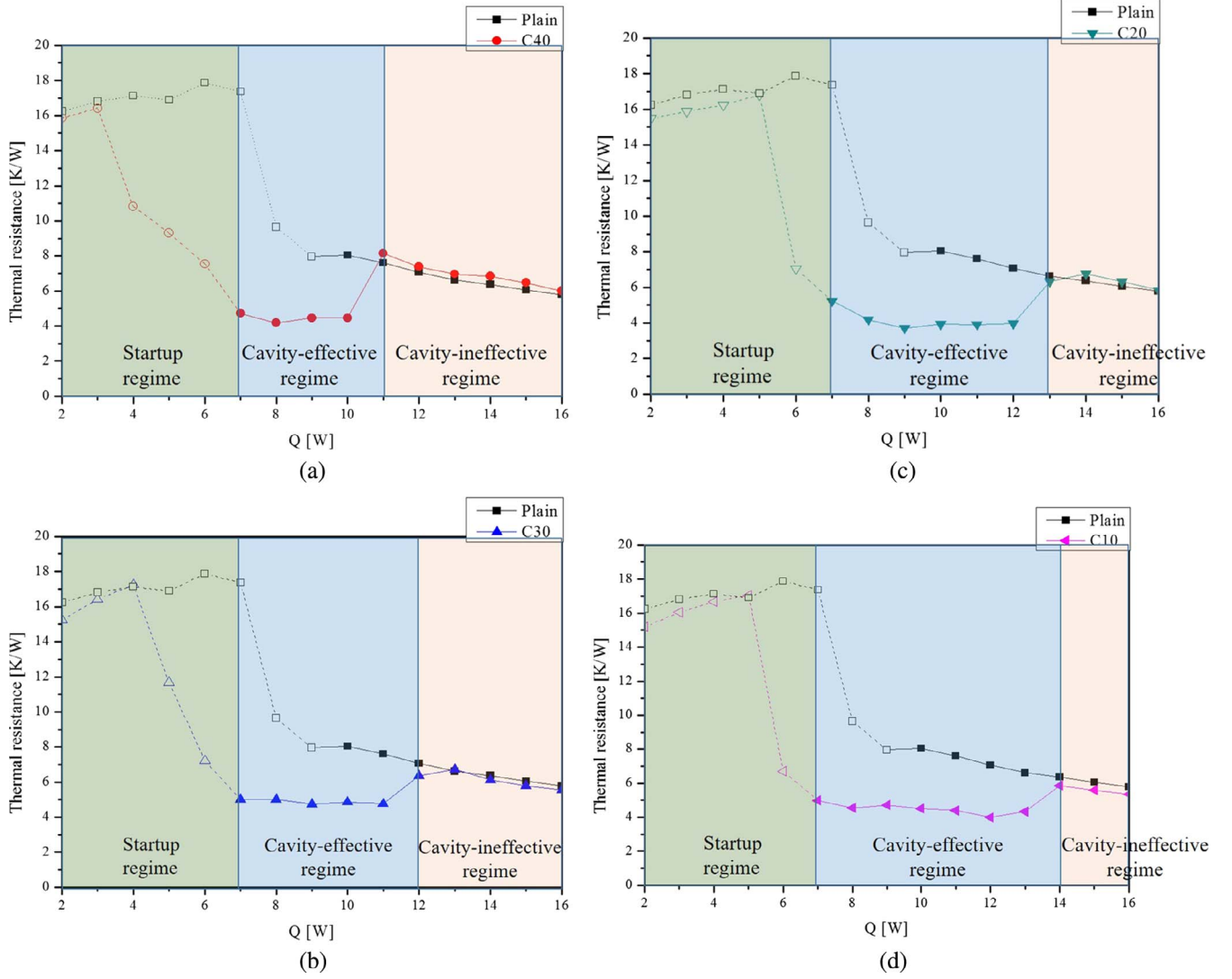


Fig. 6. Three regimes of the MPHPs with reentrant cavities (C40).

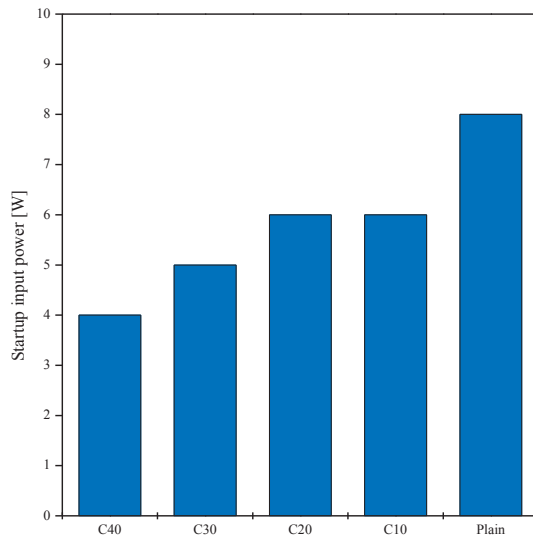


Fig. 7. Startup input power of the MPHPs with reentrant cavities and the plain MPHP.

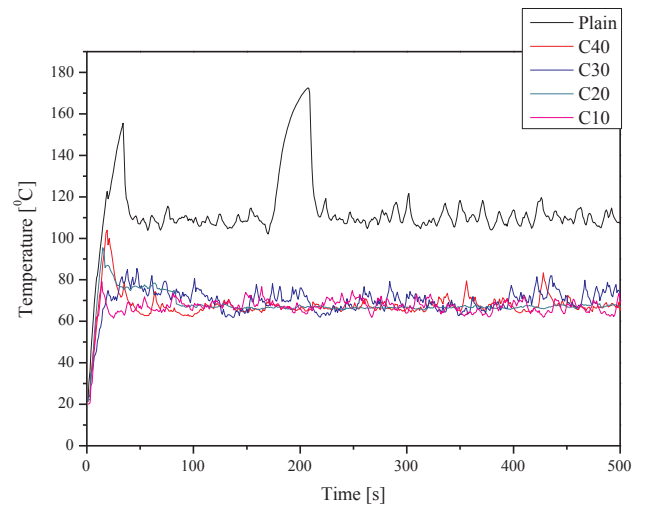


Fig. 8. The temperatures of the evaporator section of the MPHPs with reentrant cavities and the plain MPHP at 9 W.

becomes similar to the plain MPHP at lower input power. In other words, a reduction of the thermal resistance of the MPHPs is sustained up to higher input power as the size of the reentrant cavities decreases. It indicates that the cavity-effective regime is widened as the size of the reentrant cavities decreases. The case of C10 shows the lowest thermal resistance at 13 W. The physical explanation on why MPHP with reentrant cavities behaves differently depending on the size of the reentrant cavities is dealt with in Section 3.1.3.

3.1.2. Flow characteristics

Fig. 9 shows the flow behavior of the MPHP with reentrant cavities (C40) and the plain MPHP at 5 W, which is in the startup regime. There is no movement in the case of the plain MPHP. Hence, the heat is transferred from the evaporator section to the condenser section only through conduction. On the contrary, in the case of the MPHP with reentrant cavities, the required wall superheat at the ONB is reduced due to reentrant cavities and early startup occurs accompanied by nucleation. The process of startup and obvious nucleation is shown in Fig. 10. Every channel in the evaporator section is occupied with vapor except for a single channel, as shown in Fig. 10(a). Following a nucleation at fabricated reentrant cavities, drastic growth of the bubbles and movement of the liquid slugs are observed, as shown in Fig. 10(b) ~ (d). After startup, the temperature of the evaporator section decreases drastically as a result of nucleation. However, in the startup regime, intermittent stopover and unstable oscillation are observed. It seems a driving force for sustaining stable normal operation is insufficient. Additional studies on the startup regime are required for a detailed physical explanation of intermittent stopover needs.

As the input power increases, the MPHP with reentrant cavities moves into the cavity-effective regime. Fig. 11 shows the flow behavior of the MPHP with reentrant cavities (C10) and the plain MPHP at 9 W, which is in the cavity-effective regime. Two main differences between the MPHP with reentrant cavities and the plain MPHP in the cavity-effective regime have been identified. First, in the case of the plain MPHP, there are five long liquid slugs, and they oscillate due to repetitive volume expansion and compression of adjacent vapor plugs. On the contrary, in the case of the MPHP with reentrant cavities, many short vapor plugs are observed. Because of continuous bubble generation from the reentrant cavities, short vapor plugs are continuously generated. The number of short vapor plugs changes with time due to the merging into long vapor plugs. Second, the displacement of the vapor-liquid meniscus for the MPHP with reentrant cavities is significantly different from that of the plain MPHP, as shown in Fig. 12.

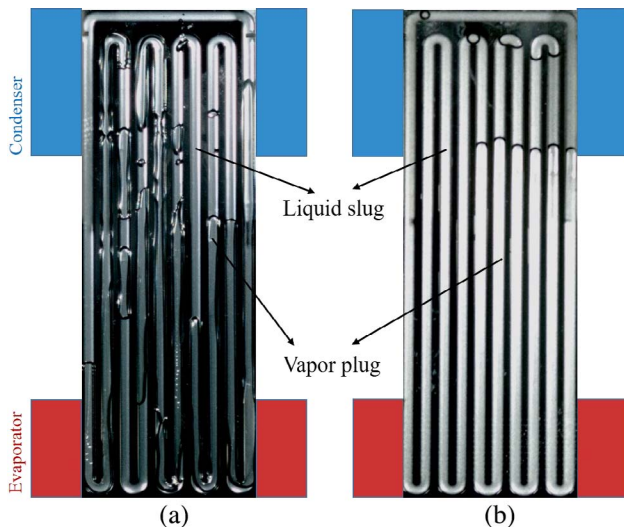


Fig. 9. Flow behavior of the MPHP in the startup regime (5 W): MPHP with reentrant cavities (C40), and (b) plain MPHP.

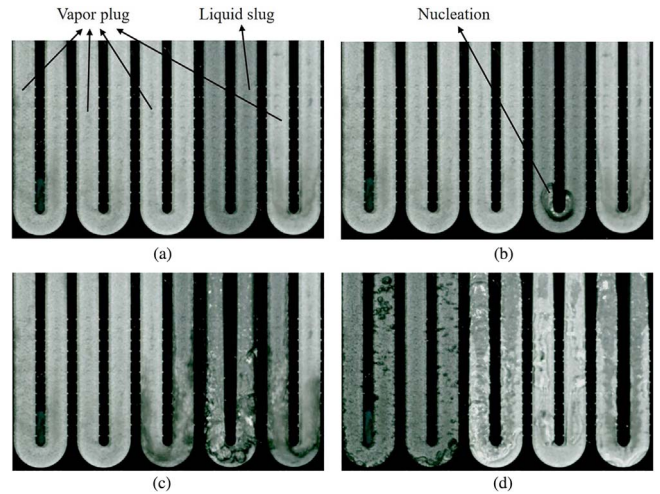


Fig. 10. Process of startup in the evaporator section of MPHP with reentrant cavities (C40): (a) 0 s, (b) 0.5 ms, (c) 3.75 ms, (d) 16.75 ms.

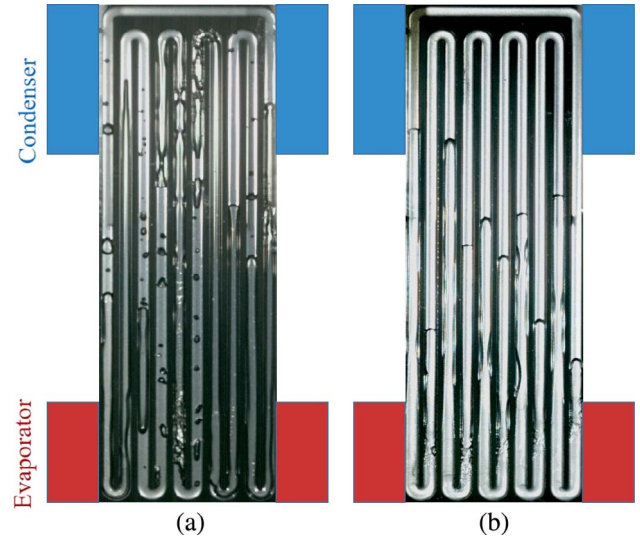


Fig. 11. Flow behavior of the MPHP in the cavity-effective regime (9 W): MPHP with reentrant cavities (C10), and (b) plain MPHP.

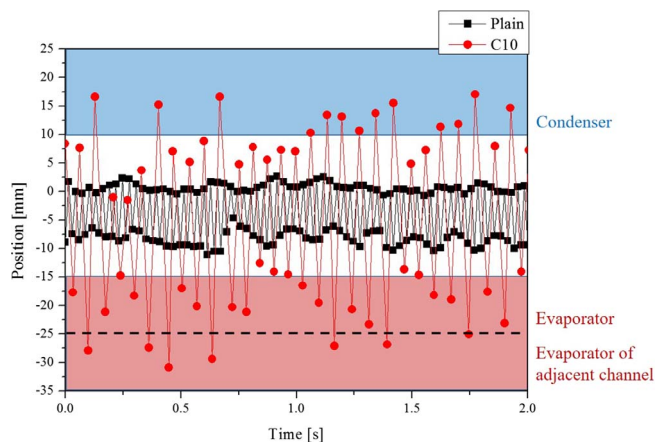


Fig. 12. Comparison of oscillation of meniscus between the MPHP with reentrant cavities (C10) and the plain PHPs with time at 9 W.

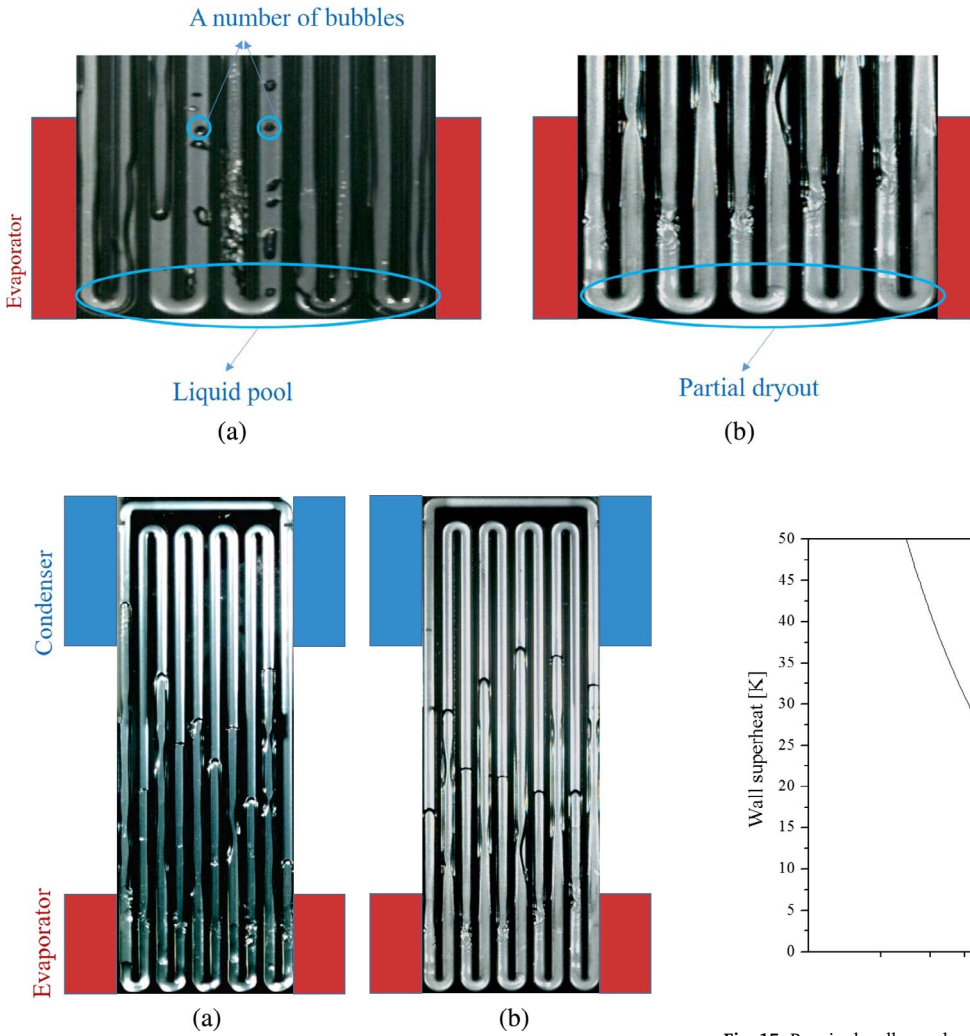


Fig. 13. Flow behavior of the evaporator section of MPHP in the cavity-effective regime (9 W) MPHP with reentrant cavities (C10), and (b) plain MPHP.

Fig. 14. Flow behavior of the MPHP in the cavity-ineffective regime (13 W) MPHP with reentrant cavities (C40), and (b) plain MPHP.

The vapor-liquid meniscus is traced to evaluate the amplitude of oscillation using flow visualization. In the case of the MPHP with reentrant cavities, the amplitude of the vapor-liquid meniscus is much larger than that of the plain MPHP. Furthermore, in the case of the MPHP with reentrant cavities, the menisci penetrate into the evaporator section more frequently compared to the case of the plain MPHP. Fig. 13 clearly shows the effects of penetration of meniscus into the evaporator section by enlarging the evaporator section of Fig. 11. In the case of the plain MPHP, partial dry out is observed, and liquid is supplied only through falling liquid film. However, in the case of the MPHP with reentrant cavities, the liquid pool in the evaporator is observed, and liquid supply is sufficient by bulk liquid motion. Therefore, the thermal resistance of the MPHPs with reentrant cavities decrease drastically. As the input power increases, the MPHP with reentrant cavities moves into the cavity-ineffective regime. The flow behavior of the MPHP with reentrant cavities and the plain MPHP in the cavity-ineffective regime are presented in Fig. 14, and there show no big difference. Furthermore, displacements of the vapor-liquid meniscus of the two show similar results.

3.1.3. Effect of the size of reentrant cavities

To analyze the effects of the size of reentrant cavities, the correlations between wall superheat at the ONB and the size of cavity, which is reported by Kandlikar et al. [24,25], is used:

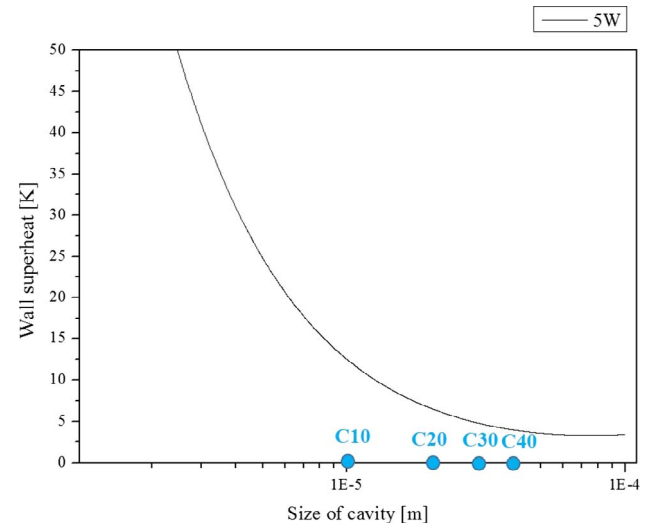


Fig. 15. Required wall superheat at the ONB for the MPHP in the startup regime (5 W).

$$\Delta T_{SAT|ONB} \text{ at } d_{cavity} = T_{WALL} - T_{SAT} = \frac{1.1 d_{cavity} q''}{2k_L \sin \theta} + \frac{4\sigma \sin \theta}{d_{cavity}} \frac{T_{SAT}}{\rho_V h_{LV}} \quad (3)$$

where $\Delta T_{SAT|ONB} \text{ at } d_{cavity}$ is the wall superheat required for ONB at the given cavity size, d_{cavity} is the size of the mouth of the cavity, q'' is heat flux at cavity, k_L is thermal conductivity of working fluid, θ is contact angle, σ is surface tension of working fluid, T_{SAT} is saturated temperature of the liquid, ρ_V is vapor density, and h_{LV} is latent heat of vaporization.

Fig. 15 presents the required wall superheat at the ONB as a function of the size of reentrant cavities in the MPHP at 5 W, which is in the startup regime. It explains why the larger size of reentrant cavities requires lower input power for promoting startup. At low input power (5 W), the required wall superheat at the ONB decreases as the size of reentrant cavities increases. Therefore, earlier startup is observed as the size of reentrant cavities increases, with the earliest startup being observed in the case of C40. Fig. 16 shows the required wall superheat at the ONB as a function of the size of the cavities in the MPHP at 9 W, which is in the cavity-effective regime. Generally, the roughness of the silicon wafer is smaller than $0.7 \mu\text{m}$ [18,26]. Compared with the roughness of silicon wafer, the size of the fabricated reentrant cavities employed in this study requires a very small amount of wall superheat in the cavity-effective regime. Hence, in the cavity-effective regime, reentrant cavities can be easily activated with lower wall superheat. Therefore, a reduction of wall superheat at the ONB induces active nucleation at reentrant cavities during normal operation and results in a drastic decrease of the thermal resistance of the MPHP with reentrant

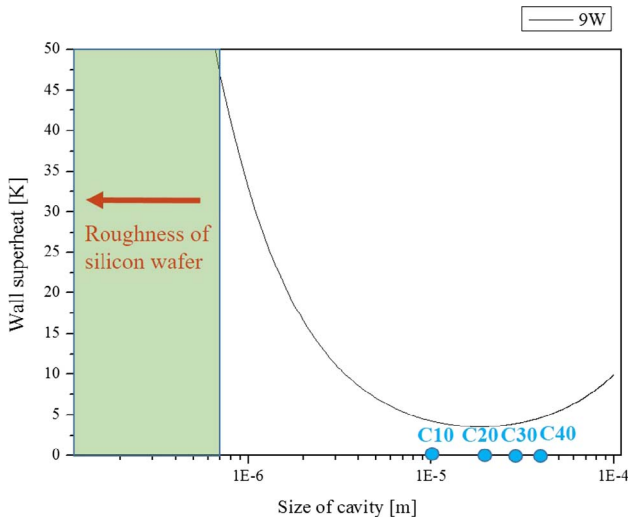


Fig. 16. Required wall superheat at the ONB for the MPHP in the cavity-effective regime (9 W).

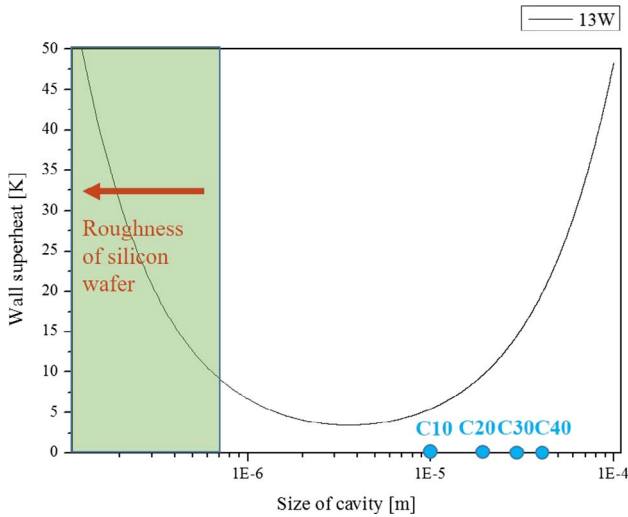


Fig. 17. Required wall superheat at the ONB for the MPHP in the cavity-ineffective regime (13 W).

cavities. As the MPHP with reentrant cavities moves into the cavity-ineffective regime, there is no difference between the thermal resistance of the MPHP with reentrant cavities and the plain MPHP. Fig. 17 clearly illustrates the required wall superheat at the ONB for the MPHP in the cavity-ineffective regime. In contrast with Fig. 16, fabricated reentrant cavities require a similar amount of wall superheat compared to the silicon wafer. Therefore, fabricated reentrant cavities have no effect on the thermal resistance of the MPHPs in the cavity-ineffective regime. As shown in Fig. 17, the required wall superheat at the ONB decreases as the size of the fabricated reentrant cavities decreases. Therefore, a reduction of the thermal resistance of the MPHPs is better maintained at higher input power as the size of the reentrant cavities decreases. The case of C10 shows the lowest thermal resistance at high input power. From Fig. 17, it can be seen that if the size of the reentrant cavities is smaller than the case of C10, a reduction of the thermal resistance of the MPHPs can be better maintained at higher input power compared to the cases conducted in this study.

Therefore, unlike the startup regime, as the size of reentrant cavities increases, the thermal resistance of the MPHP with reentrant cavities becomes similar to the plain MPHP at lower input power. Therefore, the most advantageous size of reentrant cavities varies depending on input power.

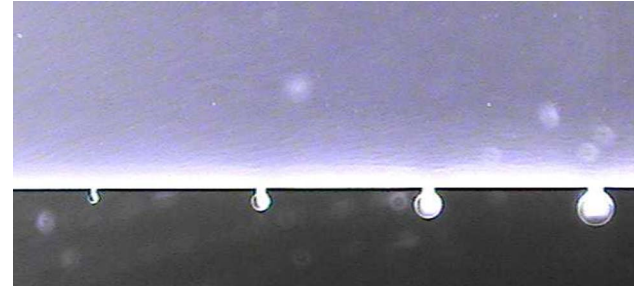


Fig. 18. Top view image of the channel of the MPHP incorporating various sizes of reentrant cavities.

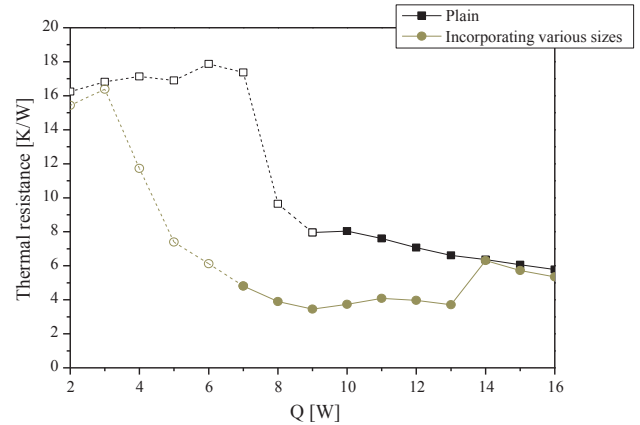


Fig. 19. The thermal resistance of the MPHP incorporating various sizes of reentrant cavities and the plain MPHP.

3.2. Optimization of thermal performance of MPHP with reentrant cavities

The effects of reentrant cavities can be optimized by incorporating various sizes of reentrant cavities into a single MPHP. Fig. 18 presents a top view image of the channel of the MPHP incorporating various sizes of reentrant cavities. Four different sizes of reentrant cavities, which are tested in this study, are employed in a single MPHP. The thermal resistance of the MPHP incorporating various sizes of reentrant cavities is evaluated using the same criterion applied in this study. The thermal resistance of the MPHP incorporating various sizes of reentrant cavities is decreased drastically compared to the plain MPHP, as shown in Fig. 19. Furthermore, the favorable effects of each size of the reentrant cavities on startup and the thermal resistance have appeared simultaneously by integrating various sizes of reentrant cavities as both earlier startup and extension of the cavity-effective regime are achieved. Considering manufacturing costs and ease of production for artificial cavities, using microporous coatings or scratch channels instead of using MEMS techniques are reasonable methods to incorporate the various sizes of cavities spontaneously. Therefore, if the range of cavity size is determined, fabricating artificial cavities with various sizes to enhance the thermal performance of PHPs, which can be done in a simple manner, is highly recommended for practical application.

4. Conclusion

In this study, the effect of the size of reentrant cavities on the thermal performance of the MPHP is investigated. Through high-speed photography and thermometry, the flow and thermal characteristics of MPHPs with reentrant cavities of different sizes are analyzed at various input powers. The MPHPs with and without reentrant cavities are fabricated onto a silicon wafer using MEMS techniques. Four different sizes of reentrant cavities, which are 10, 20, 30, 40 μm , are engraved inside the four MPHPs with each MPHP having only one size of

reentrant cavities. In the case of the MPHPs with reentrant cavities, input power required for startup and the thermal resistance are decreased by up to 50% and 57% respectively, compared to the MPHP without reentrant cavities. Reentrant cavities promote active nucleation by reducing the wall superheat required for the ONB and provide a larger driving force for the oscillating motions of the menisci. This in turn leads to early startup as well as a decrease of the thermal resistance.

Earlier startup is observed as the size of reentrant cavities increases, and the MPHP with 40 μm cavities shows the earliest startup. On the other hand, the cavity-effective regime is widened as the size of reentrant cavities decreases and the MPHP with 10 μm cavities shows the widest cavity-effective regime. These two phenomena can be explained by the fact that the most advantageous size of reentrant cavities varies with input power. Hence, the MPHP with the largest cavities (40 μm) and the MPHP with the smallest cavities (10 μm) are favorable for low and high input power, respectively. To take advantage of favorable effects of the different sizes of reentrant cavities, a single MPHP incorporating various sizes of reentrant cavities (10, 20, 30, and 40 μm) is proposed and tested. In the case of the MPHP incorporating various sizes of reentrant cavities, both early startup and extension of cavity-effective regime are accomplished simultaneously.

Acknowledgement

This work was supported by the National Research Foundation of Korea (NRF) grant which is funded by the Korean government (MISP) (No. 2012R1A3A2026427).

References

- [1] H. Akachi, F. Polasek, P. Stulc, Pulsating Heat Pipes, in: Proceedings of 5th International Heat Pipe Symposium, Melbourne, Australia (1996) 208–217.
- [2] G.H. Kwon, S.J. Kim, Experimental investigation on the thermal performance of a micro pulsating heat pipe with a dual-diameter channel, *Int. J. Heat Mass Transf.* 89 (2015) 817–828.
- [3] T. Inoue, D. Motouchi, K. Mushiga, Y. Suzuki, Oscillating heat pipe for thermal management of space satellite, in: ASME-JSME Thermal Engineering Summer Heat Transfer Conference, Vancouver, Canada, 2007.
- [4] R. Senjaya, T. Inoue, Bubble generation in oscillating heat pipe, *Appl. Therm. Eng.* 60 (2013) 251–255.
- [5] R. Senjaya, T. Inoue, Oscillating heat pipe simulation considering bubble generation part I: presentation of the model and effects of a bubble generation, *Int. J. Heat Mass Transf.* 60 (2013) 816–824.
- [6] Q.M. Li, Z. Yang, Y.Y. Duan, B.X. Wang, Temperature and flow fluctuations in a quartz-glass oscillating heat pipe, *IEEE* (2011) 209–213.
- [7] C.T. Lu, C. Pan, Convective boiling in a parallel microchannel heat sink with a diverging cross section and artificial nucleation sites, *Exp. Therm Fluid Sci.* 35 (2011) 810–815.
- [8] D.T. Pate, R.J. Jones, S.H. Bhavnani, Cavity-induced two-phase heat transfer in silicon microchannels, *IEEE* (2006) 71–78.
- [9] J.Y. Lee, M.H. Kim, M. Kaviany, S.Y. Son, Bubble nucleation in microchannel flow boiling using single artificial cavity, *Int. J. Heat Mass Transf.* 54 (2011) 5139–5148.
- [10] C.K. Yu, D.C. Lu, T.C. Cheng, Pool boiling heat transfer on artificial micro-cavity surfaces in dielectric fluid FC-72, *J. Micromech. Microeng.* 16 (2006) 2092–2099.
- [11] W. Qu, H.B. Ma, Theoretical analysis of startup of a pulsating heat pipe, *Int. J. Heat Mass Transf.* 50 (2007) 2309–2316.
- [12] N.D. Nimkar, S.H. Bhavnani, R.C. Jaeger, Effect of nucleation site spacing on the pool boiling characteristics of a structured surface, *Int. J. Heat Mass Transf.* 49 (2006) 2829–2839.
- [13] L. Dong, X. Quan, P. Cheng, An experimental investigation of enhanced pool boiling heat transfer from surfaces with micro/nano-structures, *Int. J. Heat Mass Transf.* 71 (2014) 189–196.
- [14] K.G. Rajulu, R. Kumar, B. Mohanty, H.K. Varma, Enhancement of nucleate pool boiling heat transfer coefficient by reentrant cavity surfaces, *Heat Mass Transf.* 41 (2004) 127–132.
- [15] Y.Y. Hsu, On the size range of active nucleation cavities on a heating surface, *J. Heat Transf.* 84 (1962) 207–213.
- [16] Y. Qi, J.F. Klausner, Heterogeneous nucleation with artificial cavities, *J. Heat Transf.* 127 (2005) 1189–1196.
- [17] A.K. Das, P.K. Das, P. Saha, Performance of different structured surfaces in nucleate pool boiling, *Appl. Therm. Eng.* 29 (2009) 3643–3653.
- [18] C.J. Kuo, Y. Peles, Local measurement of flow boiling in structured surface microchannels, *Int. J. Heat Mass Transf.* 50 (2007) 4513–4526.
- [19] D. Deng, Y. Tang, D. Liang, H. He, S. Yang, Flow boiling characteristics in porous heat sink with reentrant microchannels, *Int. J. Heat Mass Transf.* 70 (2014) 463–477.
- [20] F. Edelstein, R. Kosson, A high capacity re-entrant groove heat pipe for cryogenic and room temperature space applications, *Cryogenics* 32 (1992) 168–172.
- [21] S.K. Thomas, V.C. Damle, Fluid flow in axial reentrant grooves with application to heat pipes, *J. Thermophys. Heat Transf.* 19 (2005) 395–405.
- [22] Y.J. Youn, S.J. Kim, Fabrication and evaluation of a silicon-based micro pulsating heat spreader, *Sens. Actuat. A* 174 (2012) 189–197.
- [23] J.P. Holman, *Experimental methods for engineers*, seventh ed., McGraw-Hill (2001) 48–106.
- [24] S.G. Kandlikar, V. Mizo, M. Cartwright, E. Ikenze, Bubble nucleation and growth characteristics in subcooled flow boiling of water, in: ASME Proceedings of the 32nd National Heat Transfer Conference 4, HTD-vol. 342, (1997) 11–18.
- [25] S.G. Kandlikar, Nucleation characteristics and stability considerations during flow boiling in microchannels, *Exp. Therm Fluid Sci.* 30 (2006) 441–447.
- [26] L.X. Yang, Y.M. Chao, L. Jia, C.B. Li, Wettability and boiling heat transfer study of black silicon surface produced using the plasma immersion ion implantation method, *Appl. Therm. Eng.* 99 (2016) 253–261.